

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION
Question Paper

Month and Year of Examination: **May/June 2025** Branch: **IIoT**
Programme: **B.Tech** Semester: **4th**
Subject: **Industry 4.0 and IIoT**
Course No: **IIPC-211**
Number of Questions to be Attempted: **5**
Total No. of Questions: **5**

Maximum Marks: **50**
Time Allowed: **3 Hrs.**
Total No. of Pages Used: **1**

Unless stated otherwise, the symbols have their usual meanings in context with the Subject.
Assume suitably and state, additional data required, if any.
Note: All questions are compulsory. There is internal choice in Question 5. Draw proper diagrams and assume necessary information.

1. What are various industrial revolutions for IIoT ? What is Industrial Internet and why it is required ? What are smart factories ? [4+3+3]
2. Explain Wireless Sensor Nodes. Write in detail the various communication protocols used with WSN. [2+8]
3. (A) Discuss IoT Gateways in detail with diagrams. [5]
(B) Describe Cloud computing with types and services. [5]
4. (A) Explain Cyber Physical Systems with examples in detail. [6]
(B) Discuss Augmented Reality with examples. [4]
5. (A) Discuss three Industrial IoT applications in detail. [10]

OR

5. (B) Write short note on the following. [3+3+2+2]
- ✖ (i) IoT innovations in Retail (ii) Profinet (iii) VLANs (iv) Intelligent Devices

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